Baby cereals / grain products, dry (rice-based) — HMTc certification standard

Institute of Contaminant Standards — Heavy Metal Tested & Certified program

Version 1.3 · Effective 2026-09-10 · Native basis: as_sold · 8/8 totals with a published limit · plus Inorganic arsenic (iAs)

STANDARD

baby-cereals-dry-rice-based, version 1.3. HMTc Method v2.0 (government-limit default, 8:1 basis, read-across)

PROVENANCE

Compiled from data/evidence/regulatory_limits.csv. Each limit is a lookup against sovereign law, not an occurrence computation.

TAMPER-EVIDENT SEAL

The SHA-256 of this exact PDF is sealed by an RFC-3161 trusted timestamp. The token (standard.pdf.tsr) and the manifest (manifest.json) accompany this file; the date of record is the timestamping authority's, not the program's.

Abstract

This is the frozen, version-pinned Heavy Metal Tested & Certified standard for baby cereals / grain products, dry (rice-based). Each per-analyte certification limit is the strictest heavy-metal maximum level set by any government in the world for this product class, expressed in the product's native basis (as_sold) and derived by lookup against sovereign law under HMTc Method v2.0 (government-limit default, 8:1 basis, read-across). The per-jurisdiction derivation below is the complete "show your work". This snapshot is immutable: any material change to a limit mints a new version with its own timestamped snapshot.

1 | Certification limits

All values in ppb (µg/kg), as_sold basis. A brand meeting every published value on a like-for-like basis qualifies for the mark. Tier 1 toxics (Pb, Cd, tAs, mercury) carry a zero-tolerance posture; Tier 2 metals (Ni, Al, Sn, total Cr) carry the program's transitional allowance. Inorganic arsenic (iAs) is an additional species claim on rice, infant, and baby-cereal pathways; it never replaces total arsenic. Where a limit is set in another basis by the binding jurisdiction it is converted into the native basis and the conversion is named in the "Basis derivation" column. When both a species and its parent total are numeric, the published species ceiling is min(species floor, parent total) by identity (iAs ≤ tAs, MeHg ≤ tHg, Cr-VI ≤ Cr); the uncapped species instrument remains in the derivation table.

both

ANALYTE	TIER	LIMIT (PPB)	BINDING JURISDICTION	BASIS DERIVATION	RATIONALE
Lead (Pb)	Tier 1	20	European Union	face value	gov_floor

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Cadmium (Cd)	Tier 1	40	European Union	face value	gov_floor
Total arsenic (tAs)	Tier 1	50	India	face value	gov_floor
Inorganic arsenic (iAs)	Tier 1	50	US-FDA	iAs 100 µg/kg (US fda2020_ias_infant_-rice_cereal_100) capped by tAs 50 µg/kg (IN in_tas_infant-milk--substitute-and--infant-foods_50) via iAs≤tAs	gov_floor
Mercury (tHg)	Tier 1	8	Israel	face value (nearest government value; comparator vs any direct limit)	read_across
Nickel (Ni)	Tier 2	3000	European Union	face value	gov_floor
Tin (Sn)	Tier 2	5000	India	face value	gov_floor
Aluminium (Al)	Tier 2	500	Australia / NZ	face value (nearest government value; comparator vs any direct limit)	read_across
Total chromium (Cr)	Tier 2	1000	HK	face value (nearest government value; comparator vs any direct limit)	read_across

2 | Per-jurisdiction derivation

The binding value for each analyte is the lowest sovereign maximum level across the jurisdictions on record, in the native basis. Rows marked "converted" were set by the jurisdiction in the other basis and carried across the reconstitution ratio; rows marked "native" were already in this basis. Every value is keyed to its record in the sovereign register.

both

ANALYTE	JURISDICTION	LIMIT (PPB)	BASIS	SOVEREIGN REGISTER ID
Pb	Brazil	20	converted	br_pb_cereal-based-foods-for-infant-young--chil_20
—	Codex Alimentarius	20	converted	codex_pb_cereal-based-foods-for-infants--and-young_20
—	European Union-(binding)	20	converted	eu2023_pb_babyfood_cereal_20
—	HK	20	converted	hk_cap132v_pb_cereal-based-food-intended--to-be-consumed-principally-b_20
—	US-FDA	20	converted	fda2025_pb_dry_infant_cereal_20
—	United Kingdom	50	converted	uk_pb_processed-cereal-based-foods-and--baby-fo_50

—	India	200	converted	in_pb_infant-milk-substitute-and-infant--foods_200
—	MERCOSUR	200	converted	mercosur_gmc12_2011_cereals_excluding_-wheat_rice_oils_pb_200
—	PY	200	converted	py_decreto1335_2019_cereals_excluding_-wheat_rice_oils_pb_200
Cd	European Union- (binding)	40	converted	eu2023_cd_babyfood_cereal_40
—	Gulf (GSO)	40	converted	gso193_2021_cd_processed_cereal_based_-foods_and_baby_foods_for_infants_and_-young_childr_40
—	United Kingdom	40	converted	uk_cd_processed-cereal-based-foods-and--baby-fo_40
—	Brazil	50	converted	br_cd_cereal-based-foods-for-infant-young--chil_50
—	China	60	converted	cn_cd_cereal-based-complementary-foods--for-inf_60
—	India	100	converted	in_cd_infant-milk-substitute-and-infant--foods_100
—	MERCOSUR	100	converted	mercosur_gmc12_2011_cereals_excluding_-wheat_rice_oils_cd_100
—	PY	100	converted	py_decreto1335_2019_cereals_excluding_-wheat_rice_oils_cd_100
tAs	India (binding)	50	converted	in_tas_infant-milk-substitute-and-infant--foods_50
—	MERCOSUR	300	converted	mercosur_gmc12_2011_cereals_excluding_-wheat_rice_oils_tas_300
—	PY	300	converted	py_decreto1335_2019_cereals_excluding_-wheat_rice_oils_tas_300
iAs	Canada	100	converted	ca_ias_rice-based-foods-intended--specifically-f_100
—	South Korea	100	converted	kr_ias_infant-formula-follow-up-formula--cereal_100
—	US-FDA (binding)	100	converted	fda2020_ias_infant_rice_cereal_100
—	Brazil	150	converted	br_ias_cereal-based-foods-for-infant--young-chil_150
—	China	200	converted	cn_ias_cereal-based-complementary-foods--for-inf_200
Ni	European Union- (binding)	3000	converted	eu2024_ni_processed_cereal_babyfood_3000
Sn	India (binding)	5000	converted	in_sn_infant-milk-substitute-and-infant--foods_5000

3 | Method and provenance

HMTc Method v2.0 (government-limit default, 8:1 basis, read-across). The default standard for any (product, analyte) is the strictest maximum level set by a credible government regulator, converted to the row's native basis. Where the law itself splits a limit by formulation, the standard adopts the law's own split and cites it. When both a species and its parent total carry numeric claims, the published species ceiling is min(species government or read-across floor, parent-total limit), disclosed as a cap (iAs ≤ tAs, MeHg ≤ tHg, Cr-VI ≤ total Cr). A species claim is never published above its parent. Read-across limits are disclosed as read-across and never presented as direct governance.

The occurrence and toxicology corpus that contextualises these limits is maintained independently as the Heavy Metal Index literature baseline (heavymetalindex.com). This certification standard cites that baseline one way; it does not use the baseline's contents as the basis for any government-limit value stated here.

4 | Version and tamper-evidence

This is version 1.3 of the Baby cereals / grain products, dry (rice-based) standard. A certification number becomes final when its version carries a resolvable DataCite DOI; a material change to any limit mints a new version with its own frozen snapshot, so a superseded number is never silently overwritten. Tamper-evidence does not depend on trusting the program: the SHA-256 of this exact PDF was submitted to an RFC-3161 trusted timestamping authority (https://freetsa.org/tsr), which returned a signed token binding that hash to a time it — not the program — attests. The token (standard.pdf.tsr), the authority's CA chain (tsa-cacert.pem), and the manifest (manifest.json) travel with this file.

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